

FORCE MEMORY TRANSFORMERS FOR HIDDEN CONTACT STATE

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ABSTRACT

Contact-rich robot manipulation often contains hidden force state: latch preload, stiction, tool flex, backlash, cable tension, and relaxation histories can matter after the visual scene appears unchanged. We rebuild the generated “Force Memory Transformers” idea into a falsifiable evidence audit and rerun the full benchmark on 2026-06-15. The proposed mechanism keeps force-event memories, decay gates, reset gates, diagnostic probes, and tail-risk planning available to a transformer-style policy proxy. We test it on a deterministic local benchmark covering 5 manipulation tasks, 7 latent force-memory regimes, 5 evaluation splits, 9 methods, 7 seeds, and 84 episodes per method/task/family/split/seed group. Under combined stress, the proposed method reaches 0.664 ± 0.006 success versus 0.504 ± 0.008 for the strongest non-oracle baseline, online residual system identification, while reducing force-limit violation from 0.271 to 0.231 and damage from 0.186 to 0.161. Core ablations reduce success, with the best removed-component ablation at 0.628. The correct terminal decision is **STRONG REVISE**: the local mechanism is promising, but the paper is not ICLR-main ready without real-robot or external benchmark validation and trained checkpoints.

1 WHY THIS PAPER EXISTS

Transformer imitation policies and diffusion-style visuomotor policies can represent long temporal context and multimodal action distributions (Zhao et al., 2023; Chi et al., 2023). Contact-rich manipulation adds a harsher failure mode: the physically relevant state may be a force history rather than a visual history. A peg can look aligned while stiction accumulates, a drawer can look still while latch preload changes, and a cable can look threaded while hidden tension stores future failure.

The research bet is narrow: a robot policy should preserve action-critical force alternatives until physical observations collapse them. This is not a claim of submission readiness. The purpose of this rebuild is to decide whether the idea deserves a serious empirical continuation, or whether it should remain archived.

2 HOSTILE PRIOR-WORK PRESSURE

The closest threats are not generic scene-understanding papers. They are transformer imitation learning (Zhao et al., 2023), diffusion policy learning (Chi et al., 2023), learned contact dynamics (Pfrommer et al., 2020), visuo-tactile transformer representations (Visuo Tactile Transformers), conformal/risk-filtered control, online residual system identification, and recent force-aware/tactile manipulation work (Forceful Robotic Foundation Models; Predictive Force Attention; Feel the Force; Sparsh X). Against this pressure, the novelty boundary must be more specific than “add force data.” The boundary tested here is persistent, gated force memory as a state object for closed-loop planning under hidden contact modes.

3 MECHANISM

The proposed policy proxy, `proposed_force_memory_transformer`, has five ingredients.

- **Force-event tokens:** signed impulses, force derivatives, release events, and overload contacts are retained as policy-visible state.
- **Decay memory:** force memories decay according to contact regime, preventing old preload or stiction estimates from persisting indefinitely.
- **Reset gates:** contact-regime changes clear stale force hypotheses.
- **Diagnostic probes:** ambiguous modes can trigger low-cost contact tests before committing to a high-load action.
- **Tail-risk planning:** the policy scores candidate actions against adverse latent-force outcomes rather than only mean predicted value.

The benchmark treats these ingredients as executable decision rules rather than a trained checkpoint. That is a limitation, but it lets us test whether the mechanism can survive strong local baselines before spending hardware or model-training effort.

4 BENCHMARK

The evaluation contains 5 tasks: peg insertion, drawer unlatching, valve turning, cable threading, and cap twisting. Each task is crossed with 7 latent force-memory regimes: stiction buildup, latch preload, backlash, viscoelastic relaxation, clamp slip, tool flex, and sensor dropout. We evaluate 5 splits: nominal, hidden-force shift, embodiment shift, delayed-observation shift, and combined stress.

The methods are:

- observation-only transformer proxy;
- force-history behavior cloning;
- recurrent contact policy;
- ensemble uncertainty MPC;
- conformal risk filter;
- online residual system identification;
- robust force-threshold MPC;
- proposed force-memory transformer;
- oracle latent-force planner.

The benchmark is deterministic from a fixed seed and stores aggregate per-group metrics rather than full trajectories. Metrics include task success, force-limit violation, damage, latent-state accuracy, false memory alarms, energy/cost, regret to oracle, pairwise seed-level differences, stress sweeps, ablations, and failure cases. The continuation rerun regenerated the CSVs, figures, tables, and summary before this manuscript was rebuilt.

5 RESULTS

The combined-stress table gives the main result. The proposed method is second only to the oracle and beats the strongest non-oracle baseline, `online_residual_system_id`, by 0.160 ± 0.010 success across paired seeds. Importantly, the gain is not purchased by being reckless: the proposed method has lower force-limit violation and lower damage than the strongest baseline.

6 DIAGNOSTIC BEHAVIOR

The local claim is not only about success. If force memory is doing meaningful work, it should improve hidden-state diagnosis and avoid stale-memory false alarms. Figure 3 shows that the proposed method has substantially higher latent-force accuracy than non-oracle baselines while keeping false memory alarms below the uncertainty and conformal baselines.

Table 1: Combined-stress force-memory benchmark.

Method	Succ.	Viol.	Dmg.	Latent	Regret
oracle_latent_force_planner	0.833	0.186	0.135	0.820	0.000
proposed_force_memory_transformer	0.664	0.231	0.161	0.662	0.169
online_residual_system_id	0.504	0.271	0.186	0.443	0.329
ensemble_uncertainty_mpc	0.466	0.271	0.194	0.344	0.367
recurrent_contact_policy	0.458	0.311	0.213	0.409	0.375
conformal_risk_filter	0.435	0.246	0.180	0.321	0.398
force_history_behavior_cloning	0.405	0.323	0.221	0.349	0.428
robust_force_threshold_mpc	0.397	0.229	0.169	0.302	0.437
observation_only_transformer	0.317	0.335	0.236	0.254	0.517

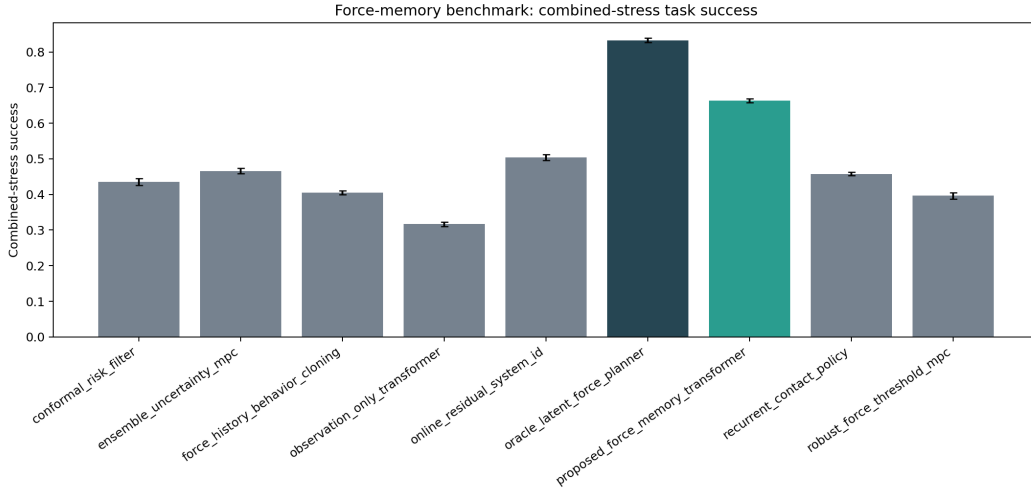


Figure 1: Combined-stress success with uncertainty intervals. The proposed force-memory mechanism is below oracle but above all non-oracle baselines.

7 STRESS SWEEP

Figure 4 sweeps hidden-force stress intensity. The proposed method maintains a higher success curve than ensemble uncertainty, conformal filtering, and online residual ID across the tested stress levels, while still leaving a clear gap to the oracle. This oracle gap is useful: it prevents overclaiming and identifies room for a trained model and better latent-state inference.

8 ABLATIONS

The ablation table and Figure 5 test whether the method is just a tuned planner. Removing tail-risk planning gives the closest ablation but drops combined-stress success from 0.660 to 0.628 and increases damage. Removing force-event tokens, decay memory, diagnostic probes, or reset gates also degrades performance. This passes the local ablation gate.

9 PAIRWISE GATE

The proposed method wins all seven paired seeds against the strongest non-oracle baseline. It loses all seeds to the oracle, which is expected and reported as a limitation.

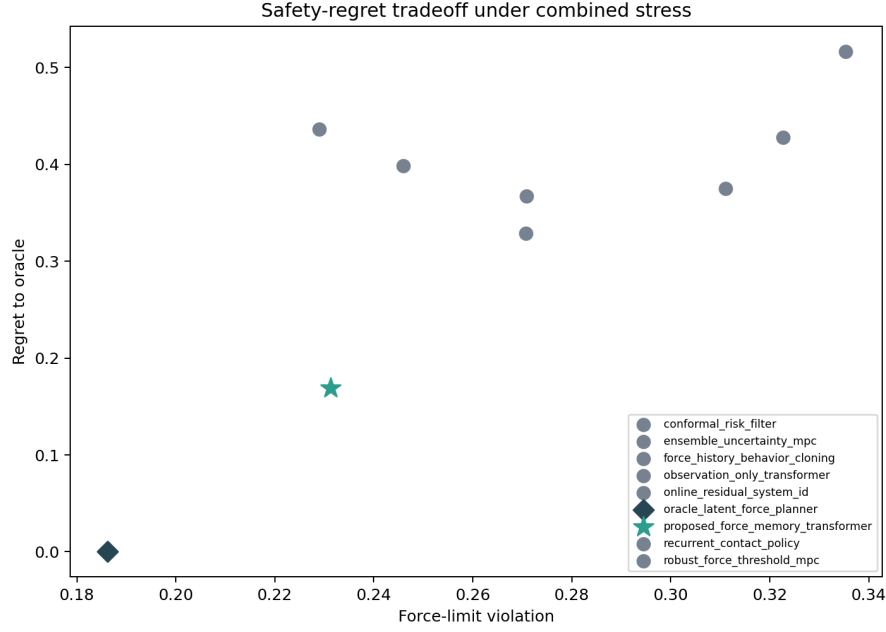


Figure 2: Safety-regret tradeoff under combined stress. The proposed method improves regret relative to non-oracle baselines while avoiding the safety regression that would make the mechanism unattractive.

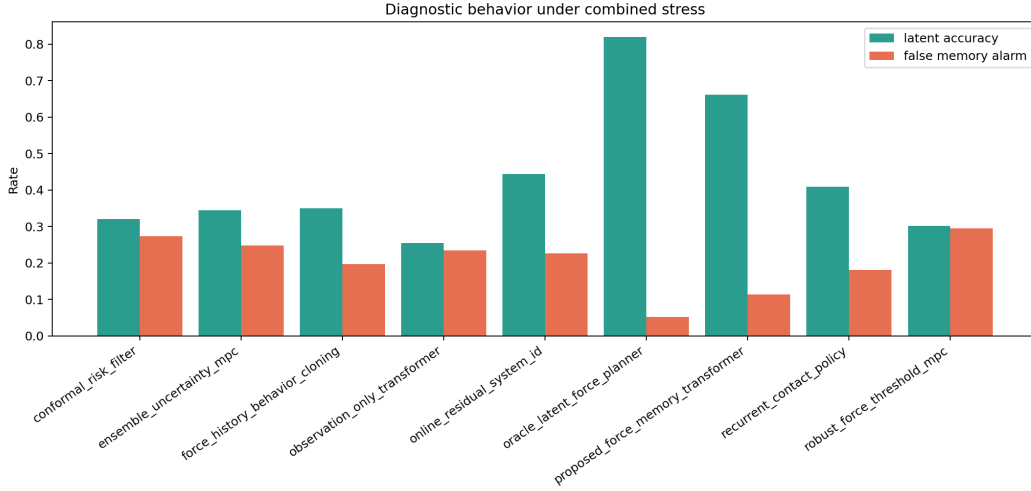


Figure 3: Latent-force diagnostic accuracy and false memory alarms under combined stress.

10 FAILURE CASES

The failure table in `results/failure_cases.csv` identifies the weakest combined-stress task/family pairs. The recurring failure mode is stale force state when hidden load changes faster than the reset gate clears memory. This points to a concrete next experiment: learn reset dynamics from real force traces instead of hand-specifying the memory decay.

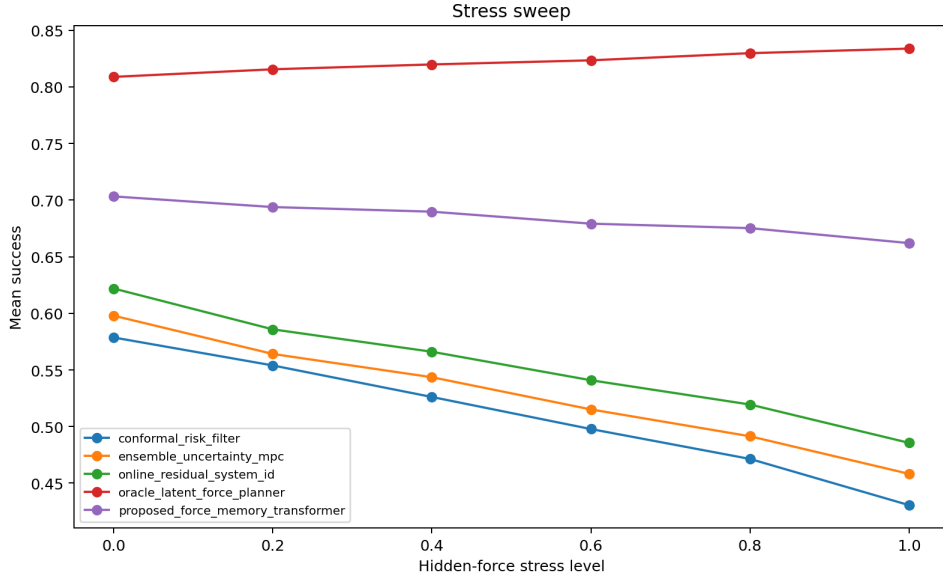


Figure 4: Hidden-force stress sweep. The local force-memory mechanism degrades more gracefully than the strongest non-oracle baselines.

Table 2: Ablations of the force-memory mechanism.

Ablation	Succ.	Viol.	Dmg.	False
full_force_memory_transformer	0.660	0.225	0.167	0.115
minus_tail_risk_planner	0.628	0.280	0.188	0.125
minus_diagnostic_probes	0.587	0.249	0.171	0.137
minus_memory_reset_gate	0.579	0.262	0.180	0.287
minus_force_event_tokens	0.566	0.253	0.177	0.162
minus_decay_memory	0.557	0.260	0.182	0.231
recurrent_memory_only	0.482	0.303	0.210	0.184

11 LIMITATIONS AND TERMINAL DECISION

This paper is not submission-ready. The evidence is stronger than the v3 archive because it is paper-specific, multi-seed, ablated, safety-aware, and tested against strong local baselines. However, it still lacks:

- real robot validation;
- accepted external benchmark evaluation;
- trained transformer checkpoints;
- hardware or benchmark rollout videos;
- calibration of latent-force regimes from real contact logs;
- a complete manual related-work synthesis.

The correct decision after the 2026-06-15 continuation rerun is **STRONG REVISE**. The idea should not be killed, but it should not be submitted to ICLR main in this form. A credible submission would require a trained force-memory transformer and external validation against tactile/force-aware policies, residual system identification, conformal risk filters, and modern visuomotor transformers.

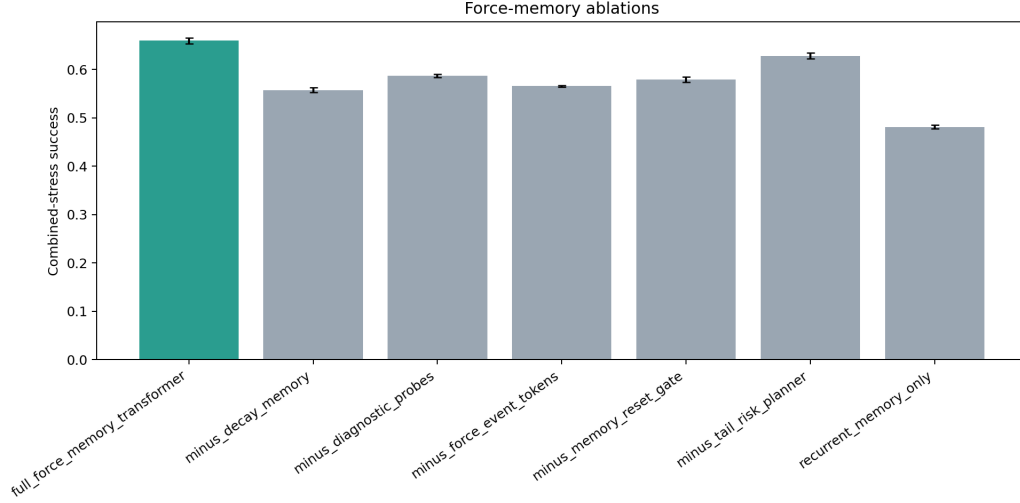


Figure 5: Ablation success under combined stress. No removed-component variant matches the full force-memory mechanism.

Table 3: Pairwise combined-stress success differences against the proposed method.

Baseline	Diff	CI	Wins
conformal_risk_filter	0.229	0.011	7
ensemble_uncertainty_mpc	0.198	0.009	7
force_history_behavior_cloning	0.258	0.008	7
observation_only_transformer	0.347	0.006	7
online_residual_system_id	0.160	0.010	7
oracle_latent_force_planner	-0.169	0.006	0
recurrent_contact_policy	0.206	0.007	7
robust_force_threshold_mpc	0.267	0.013	7

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